Mathematical Physics I (Fall 2026): Homework #4

Due Nov. 6, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

• By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code, and the SNU AI Guideline!)

• Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.

• Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-6. Boas Chapter 7, Problems 5.3, 6.12 (for Problem 5.9), 7.11, 8.15(b), 11.8, 12.24

(Note: For Problem 6.12, you will first have to work out Problem 5.9. You are asked to tackle Problem 5.9 in two ways — first directly find the Fourier coefficients, then verify your result using the answer given in Problem 5.7. Then for Problem 6.12 you may consider a way to automatically draw several partial sums of different n's — e.g., a simple script written in MATLAB or in `python`; however, making such an automated script is not necessary for you to receive a full credit. For Problem 11.8, you are asked to first work out Problem 9.10.)

7.-8. Boas Chapter 8, Problems 2.6, 3.14

(Note: For Problem 2.6, read the instruction in the textbook carefully; that is, you have been asked to plot a slope field and the solution curve with a computer. To learn more about the "slope field", you may review Boas Chapter 8, Section 1. For Problem 3.14, read the instruction and the hint in the textbook carefully.)

9. In one of the methods to convert AC voltage into DC voltage, a *full wave diode bridge rectifier* yields the waveform $V(t) = V_0 \left|\sin \omega_0 t\right|$ — i.e., passes the positive peaks of the incoming sine wave but inverts the negative peaks.

(a) After briefly discussing how a full-wave diode bridge rectifier works, expand the rectified sine wave voltage $V(t)$ in an appropriate Fourier series. Show that the lowest (fundamental) oscillation frequency is $2\omega_0$, and explain physically why all odd harmonics are completely absent.

(b) Suppose this rectified voltage signal $V(t)$ is applied across a resistor R. Calculate the total time-averaged electrical power $\overline{P_{\text{total}}}$ dissipated in R directly using the time-domain integration:

$$\overline{P_{\text{total}}} = \frac{1}{T}\int_0^T \frac{[V(t)]^2}{R}\, dt.$$

Then use Parseval's Theorem to express $\overline{P_{\text{total}}}$ as the sum of powers contributed by the DC component and each individual AC harmonic:

$$\overline{P_{\text{total}}} = P_{\text{DC}} + \sum_{n=1}^{\infty} P_n.$$

Calculate the ratio of the power contained in the DC component (P_{DC}) to the total power. Additionally, explain briefly how modern rectifier designs reduce the power remaining in unwanted AC harmonics.

(c) Using the Fourier series found above, prove that:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \frac{1}{7\cdot 9} + \cdots = \sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \frac{1}{2},$$

$$\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \frac{1}{7\cdot 9} + \cdots = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{4k^2-1} = \frac{\pi-2}{4}, \quad \text{and}$$

$$\frac{1}{1^2\cdot 3^2} + \frac{1}{3^2\cdot 5^2} + \frac{1}{5^2\cdot 7^2} + \frac{1}{7^2\cdot 9^2} + \cdots = \sum_{k=1}^{\infty} \frac{1}{(4k^2-1)^2} = \frac{\pi^2-8}{16}.$$

(Note: To learn more about how various types of rectifiers work in electronic circuits, see e.g., *The Art of Electronics* by Horowitz & Hill. For the last question in (b), you may find the following keywords useful: multi-phase rectifier, smoothing reservoir capacitor, low-pass filter, among others. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)

10. Let us use the Fourier transform to appreciate the meaning of the Heisenberg uncertainty principle in quantum mechanics.

(a) Imagine an infinite wave train $\sin \omega_0 t$ clipped by shutters to maintain only N cycles of the original waveform:

$$f(t) = \begin{cases} \sin \omega_0 t, & \text{if } |\omega_0 t| < N\pi, \\ 0, & \text{if } |\omega_0 t| > N\pi. \end{cases}$$

Find the amplitude function of the Fourier (exponential) transform, $g(\omega)$. Since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(b) Find the amplitude function of the Fourier sine transform, $g_s(\omega)$. Again, since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(c) Show that, in the special case of $N = \frac{1}{2}$ and $\omega_0 = 1$, $g_s(\omega)$ becomes

$$g_{s,1}(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega \cos\left(\frac{\omega\pi}{2}\right)}{1-\omega^2}.$$

(d) Now consider the limit of $\omega_0 \gg 1$ and $\omega \approx \omega_0$. Show that $g_s(\omega)$ is approximately equal to

$$g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{\sin[(\omega_0 - \omega)\frac{N\pi}{\omega_0}]}{\omega_0 - \omega}.$$

Sketch or computer plot this function for $N = 1, 3, 5, 10$ and an arbitrary ω_0. Notice that for $N \gg 1$, $g_{s,2}(\omega)$ may be interpreted as proportional to what we later define as the Dirac delta function, $\delta(\omega - \omega_0)$, in Boas Chapter 8, Section 11.

(Note: You do not need to provide a mathematically rigorous proof that $g_{s,2}(\omega)$ indeed becomes proportional to $\delta(\omega - \omega_0)$ as $N \to \infty$. For now, observe the shape of $g_{s,2}(\omega)$ as you vary N, and based on your observation simply argue that $\lim_{\omega \to \omega_0} g_{s,2}(\omega) \to \infty$ as $N \to \infty$.)

(e) Show that the first zeros of $g_{s,2}(\omega)$ from $\omega = \omega_0$ are at $\omega = \omega_0 \pm \Delta\omega = \omega_0 \pm \frac{\omega_0}{N}$. Justify that $\Delta\omega = \frac{\omega_0}{N}$ could be a good measure of the spread (or uncertainty) in frequency of our clipped wave train. Then, establish the inverse relationship between the wave train's pulse length ($N\pi$) and the frequency spread ($\Delta\omega$). Finally, using the relationship along with the assumed *wave nature of matter*, explain the uncertainty principle of quantum mechanics.

(Note: You may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Note that the inverse relationship found here is a fundamental property of the finite wave train, and has little to do with any additional *ad hoc* postulates in quantum mechanics. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)